COVID-19 GLOBAL IMPACT DASHBOARD

Executive Summary | Historical country-wise and time-series analysis

Project focus	Tools	Source
Cases, deaths, recoveries and active cases	Python, Pandas, Matplotlib, Power BI	Kaggle – COVID-19 Dataset (Devakumar K. P.)

1. Project Objective

To explore historical COVID-19 data across countries and dates, understand how reported cases, deaths and recoveries changed over time, compare country-level impact, and present the results in an interactive Power BI dashboard for clear, non-technical communication.

2. Dataset Description

The project uses the Kaggle **COVID-19 Dataset** by Devakumar K. P., using the **full_grouped.csv** file. It contains date-wise country/region records with fields such as Confirmed, Deaths, Recovered, Active, New cases, New deaths and New recovered. This is a historical dataset; its figures should not be presented as current COVID-19 totals.

3. Key Findings from the Analysis

- **Country comparison:** Reported case counts differ substantially between countries, so country-wise ranking helps identify where the recorded burden was highest in the selected period.
- **Time trend:** Daily new-case values vary by date; a time-series chart makes increases, decreases and periods of change easier to identify.
- **Multiple impact measures:** Confirmed cases alone do not describe the full impact. Deaths, recoveries and active cases provide additional context.
- **Fair interpretation:** Raw totals are affected by country population and reporting practices. Per-capita measures and data-quality checks are useful for fairer comparisons.

4. Recommendations

- Compare countries using rates per million people as well as raw totals, where population data is available.
- Use the latest available date in the file for cumulative country totals; do not add cumulative daily totals across dates.
- Check missing values, duplicate rows, date formats and unusual values before interpreting charts.
- Treat reported figures as estimates subject to differences in testing, reporting and dataset coverage.

5. Conclusion

The project demonstrates an end-to-end data analytics workflow: dataset preparation, exploratory analysis, visualization and dashboard reporting. The dashboard supports country and date comparisons and helps communicate historical COVID-19 patterns clearly.

Reporting note: This summary intentionally avoids unverified numeric claims. Before final submission, update the findings with the exact values and country rankings produced by the cleaned dataset and the corrected Power BI measures.